

Evaluation of Classification and Regression Models

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Evaluation and Credibility Issues



- Which measures should we use?
- How reliable are the predicted results?
- How much should we believe in what was learned?
 - Error on the training data is not a good indicator of performance on future data
 - The classifier was computed from the very same training data, any estimate based on that data will be optimistic
 - In addition, new data will probably not be exactly the same as the training data!

Model Evaluation



- Metrics for Performance Evaluation
 - How to evaluate the performance of a model?
- Methods for Performance Evaluation
 - How to obtain reliable estimates?
- Methods for Model Comparison
 - How to compare the performance of competing models?
- Model Selection
 - Which model should we prefer?



Metrics for Performance Evaluation (Classification)

- Focus on the predictive capability of a model
- Confusion Matrix:

		PREDICTED CLASS	
		P	N
ACTUAL CLASS	P	TP	FN
	N	FP	TN

TP: True Positive, TN: True Negative
FN: False Negative, FP: False Positive

- Most widely-used metrics:
 - Accuracy
 - Precision
 - Recall
 - F1-measure
 - AUC (Area under the ROC curve)



Accuracy

- Confusion Matrix:

		PREDICTED CLASS	
		P	N
ACTUAL CLASS	P	TP	FN
	N	FP	TN

$$Accuracy = \frac{TP+TN}{TP+FN+FP+TN}$$

No. of correct
predictions

- Sometimes accuracy is not enough.
- Consider a 2-class problem:
 - Number of Class 0 examples = 9990
 - Number of Class 1 examples = 10
- If model predicts everything to be class 0, accuracy is $9990/10000 = 99.9\%$
- Accuracy is misleading because model does not detect any class 1 example

Precision and Recall



- Alternatives to accuracy, introduced in the area of information retrieval and search engine
- Precision:
 - Focuses on the percentage of examples that have been classified positive and are actually positive
 - In the information retrieval context, precision represents the percentage of actually good documents that have been shown as a result
- Recall:
 - Focuses on the percentage of positively classified examples with respect to the number of existing good documents
 - In the information retrieval context, recall represents the percentage of good documents shown with respect to the existing ones.

Precision, Recall and F1-measure


$$\text{Precision } (p) = \frac{TP}{TP+FP}$$

The higher the precision,
the lower the FPs

$$\text{Recall } (r) = \frac{TP}{TP+FN}$$

The higher the recall, the
lower the FNs

$$\text{F1-measure} = \frac{2rp}{r+p}$$

- Precision is biased towards - ?
- Recall is biased towards - ?
- If the model makes a lot of FP, should we use F1-measure?



AUC (Area under the ROC curve)

ROC (Receiver Operating Characteristic) Curve

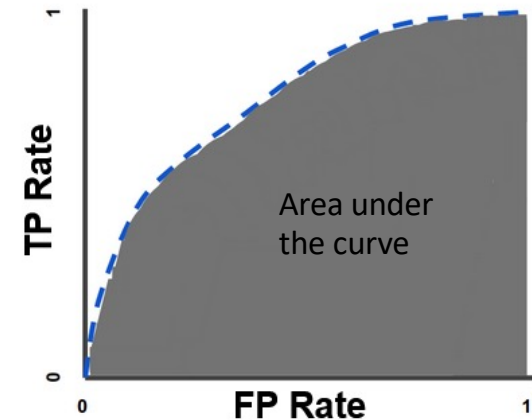
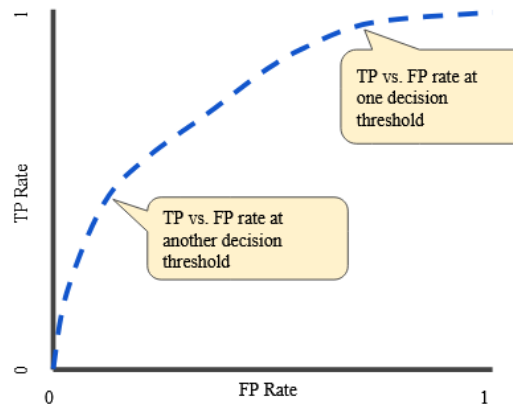
- Graph shows the performance of a classification model at all classification thresholds.
- This curve plots two parameters:
 - True Positive Rate (TPR)

$$TPR = \frac{TP}{TP+FN}$$

- False Positive Rate (FPR)

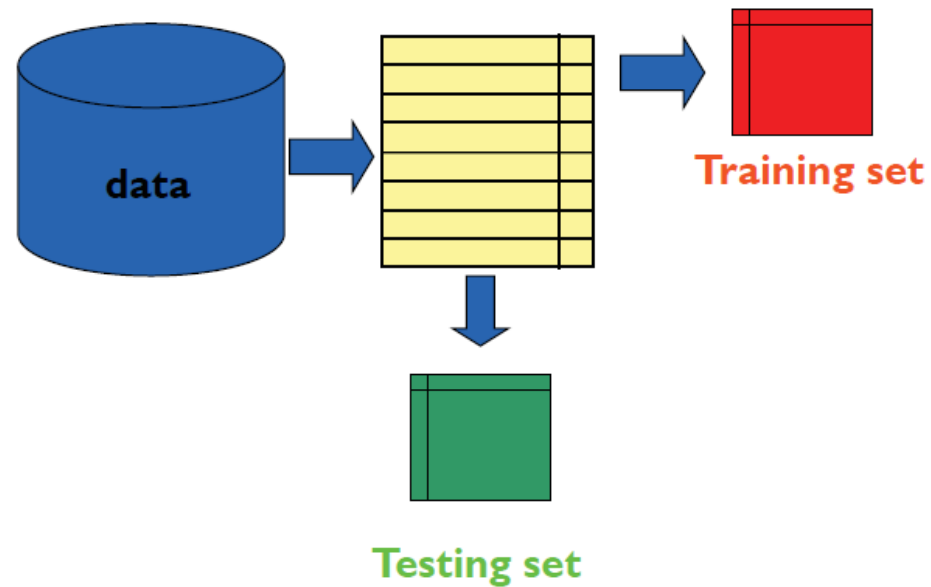
$$FPR = \frac{FP}{FP+TN}$$

***Sensitivity and specificity*



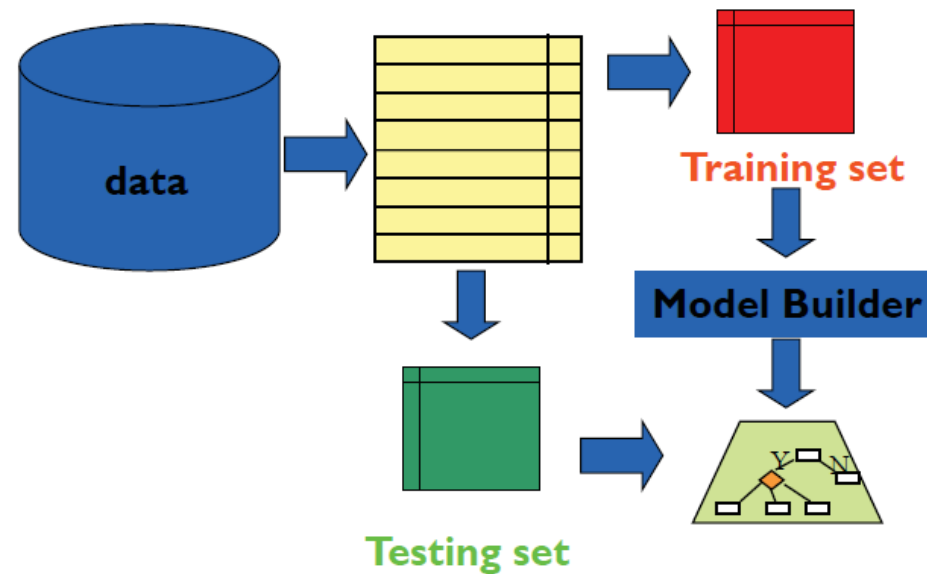
Methods for Performance Evaluation

- **Classification Step I: Split the Data Between Train and Test Set**



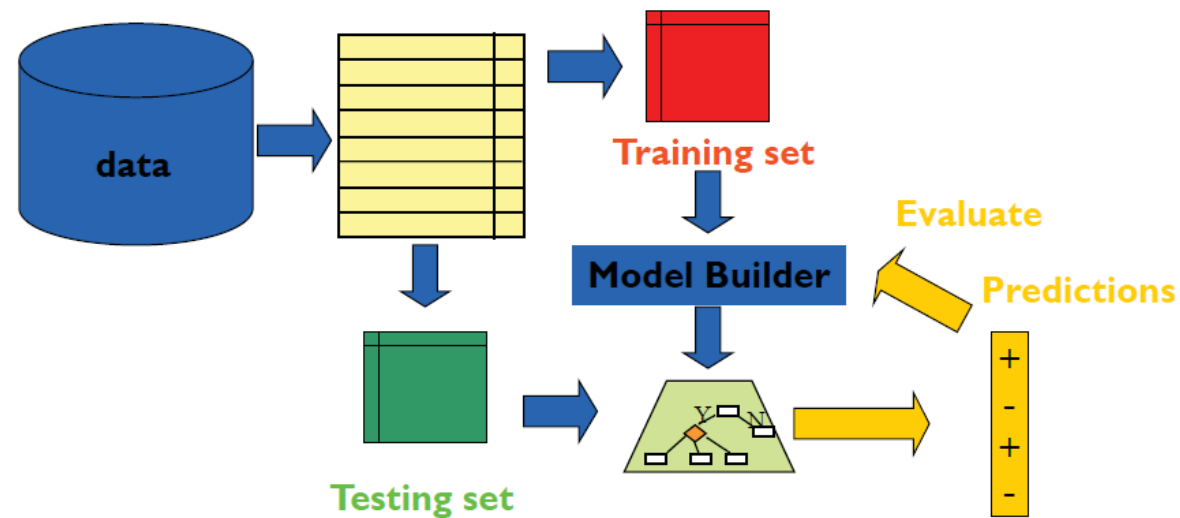
Methods for Performance Evaluation

- Classification Step 2: Compute a Model using the Data in the Training Set



Methods for Performance Evaluation

- **Classification Step 3: Evaluate the Model using the Test Set**



Note on Parameter Tuning



- It is important that the test data is not used in any way to create the classifier
- Some learning schemes operate in two stages:
 - Stage 1: builds the basic structure
 - Stage 2: optimizes parameter settings
- The test data can't be used for parameter tuning!
- Proper procedure uses three sets: training data, validation data, and test data. Validation data is used to optimize parameters

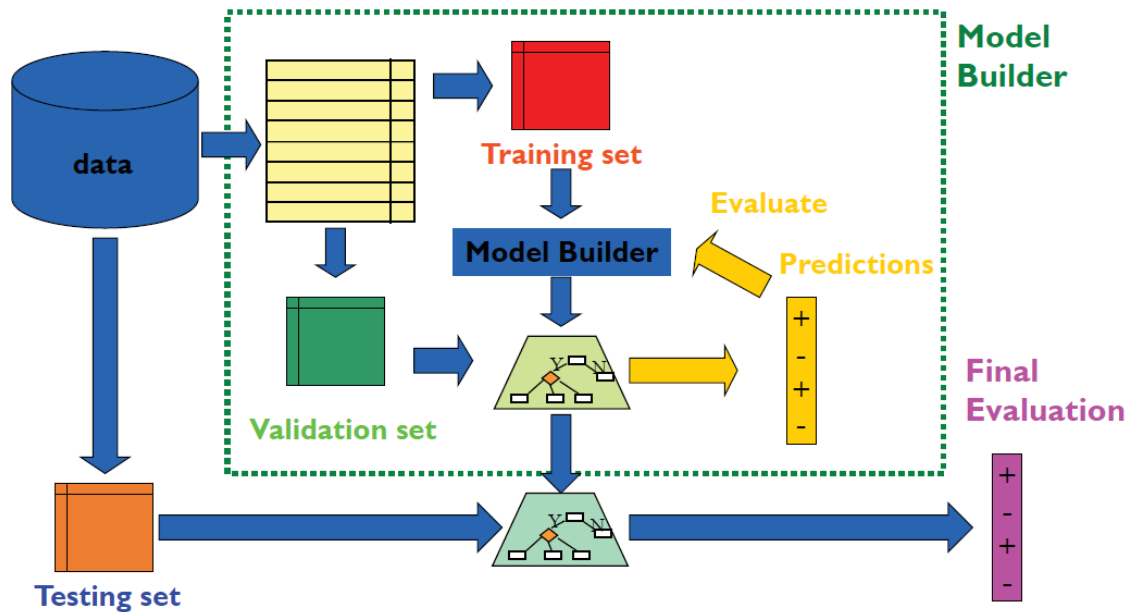
Making the most of the data



- Once evaluation is complete, all the data can be used to build the final classifier
- Generally, the larger the training data the better the classifier
- The larger the test data the more accurate the error estimate

Methods for Performance Evaluation

- The Full Process: Training, Validation, and Testing



Methods of Estimation



- **Holdout**
 - Reserve 1/2 for training and 1/2 for testing
 - Reserve 2/3 for training and 1/3 for testing
- **Random subsampling**
 - Repeated holdout
- **Stratified sampling**
 - oversampling vs undersampling
- **Cross-validation**
 - Partition data into k disjoint subsets
 - k -fold: train on $k-1$ partitions, test on the remaining one
 - Leave-one-out: $k=n$
- **Bootstrap**
 - Sampling with replacement

Holdout



- Reserves a certain amount for testing and uses the remainder for training,

Label	A	A	A	A	B	A	A	B	A	B
Training								Testing		

- For small or “unbalanced” datasets, samples might not be representative
 - For instance, it might generate training or testing datasets with few, or none instances of some classes

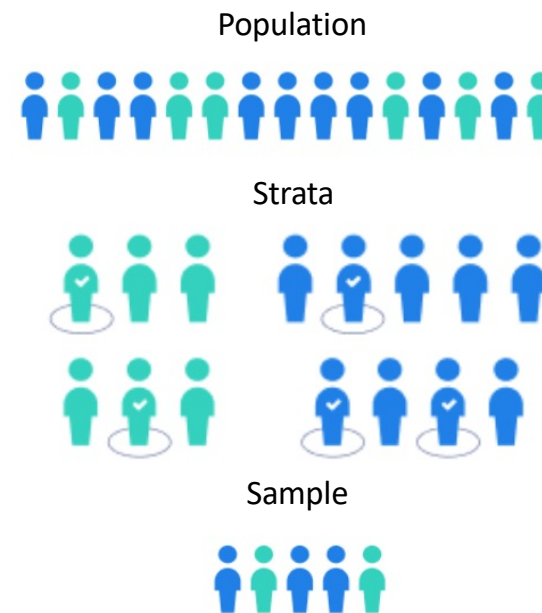
Repeated Holdout

- Holdout estimate can be made more reliable by repeating the process with different subsamples
- In each iteration, a certain proportion is randomly selected for training
- The error rates on the different iterations are averaged to yield an overall error rate

Training					Testing					
Label	A	A	A	A	B	A	A	B	A	B
Label	A	A	A	A	B	A	A	B	A	B
Label	A	A	A	A	B	A	A	B	A	B
Label	A	A	A	A	B	A	A	B	A	B

Stratified Sampling

- Sampling is conducted separately for each strata
- Makes sure that each class is represented with approximately equal proportions in both subsets



Stratified Sampling



- Oversampling method increases the minority class samples to reduce the degree of uneven distribution.
- Undersampling removes the samples from the majority class
- Imbalanced-learn python library

Bootstrapping

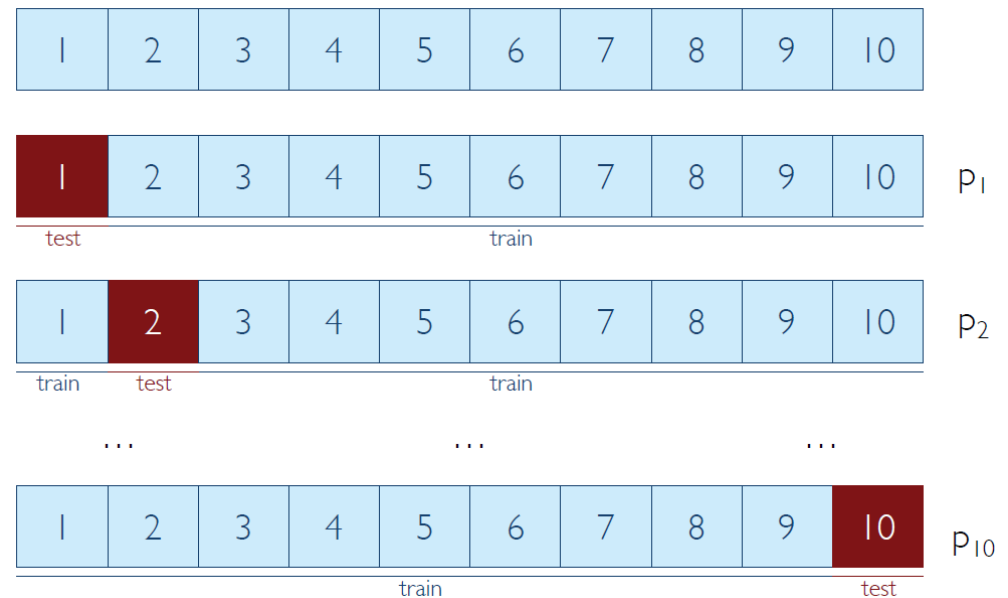
- Sampling with replacement



Cross-validation

- First step:
 - Data is split into k subsets of equal size
- Second step:
 - Each subset in turn is used for testing and the remainder for training
- This is called k -fold cross-validation and avoids overlapping test sets
- The error estimates are averaged to yield an overall error estimate

Ten-fold cross-validation



The final performance is computed as the average p_i

Cross-validation



- Standard method for evaluation stratified ten-fold cross-validation
- Why ten? Extensive experiments have shown that this is the best choice to get an accurate estimate
- Stratification reduces the estimate's variance
- Even better: repeated stratified cross-validation
- E.g., ten-fold cross-validation is repeated ten times and results are averaged (reduces the variance)
- Other approaches appear to be robust, e.g., 5x2 cross-validation

Cross-validation



Leave-One-Out Cross-Validation

- It is a particular form of cross-validation
 - Set number of folds to number of training instances
 - I.e., for n training instances, build classifier n times
- Makes best use of the data
- Involves no random subsampling
- Computationally expensive

Methods for Model Comparison



How to Compare the Performance of Two Models?

- Two models,
 - Model M1: accuracy = 85%, tested on 30 instances
 - Model M2: accuracy = 75%, tested on 5000 instances
- Can we say M1 is better than M2?
- How much confidence can we place on accuracy of M1 and M2?
- Can the difference in performance measure be explained as a result of random fluctuations in the test set?

Methods for Model Comparison



Comparing Data Mining Schemes

- Which of two learning schemes performs better?
- The answer is obviously domain-dependent
- The typical approach is to compare 10-fold cross-validation (CV) estimates using the Student's t-test
- Student's t-test tells whether the means of two samples are significantly different
- Use a paired t-test when the individual samples are paired, that is, when the same CV is applied twice. Use unpaired t-test otherwise
- Statistical Significance Test

Performance Evaluation Metrics (Regression)

- Mean absolute error (MAE)
- Mean squared error (MSE)
- Root mean squared error (RMSE)
- R^2 coefficient

$\hat{y}_i \longrightarrow$ Predicted values

$y_i \longrightarrow$ Observed values

$\bar{y}_i \longrightarrow$ Average of observed values

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|$$

$$MSE = \sum_{i=1}^n \frac{(\hat{y}_i - y_i)^2}{n}$$

$$RMSE = \sqrt{\sum_{i=1}^n \frac{(\hat{y}_i - y_i)^2}{n}}$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{\sum_{i=1}^n (\bar{y}_i - y_i)^2}$$

Useful Links



- <https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc>
- https://scikit-learn.org/stable/modules/model_evaluation.html
- https://imbalanced-learn.org/stable/under_sampling.html
- https://pandas.pydata.org/docs/user_guide/index.html#user-guide

Thank you

